

Assignment 04

Task 1(a)

In this task first i created an empty list in the size of the first input. After that I took the edges and put them in the matrix. After that i used a loop to print the values of matrix

Task 1(b)

In this task we took an empty list and appended the edges. After That I wrote the vertex of the list in the output file using a loop to run through the whole matrix

Task 2

Here at first I enqueued the vertex then explored it after exploring I dequeued that and enqueued new vertex to explore and continued this process. After that I built the graph by appending the edge and completed the task.

Task 3

For this task we are using the stack to do depth breadth search. I am pushing the values in the stack as long as it's able to go down. After that I'm back tracking the value to explore new vertices. After exploring the values we are popping the values out and working with the new values. After the whole process I'm appending the edges to the graph and creating it.

Task 4

In this task at first I took a recursive stack and a variable to check if it's already visited or not. After that I check if the circle contains any circle by creating a loop and checking the cycle vertex's and returning YES/NO on the basis of that. After that I created a graph by appending the edges to the graph.

Task 5

In this task at first I used a queue to get the possible outcomes. Then I used a while loop to get the shortest path.

Task 6

In this task I used the DFS method to solve it. I checked the current position and if there is any obstacle. I counted the number of diamonds and increased it if I got any. And followed all the instructions and conditions to get the maximum number of diamonds possible.

Task 7

In this task I did DFS on the given graph. I started from the root(city). While doing DFS, I kept track of the farthest city from the root(city A). then I started another DFS from city a to find the farthest city from it. The path from city a to city b represents the longest distance in the tree which ensures that dora can pass through the maximum number of cities exactly once.